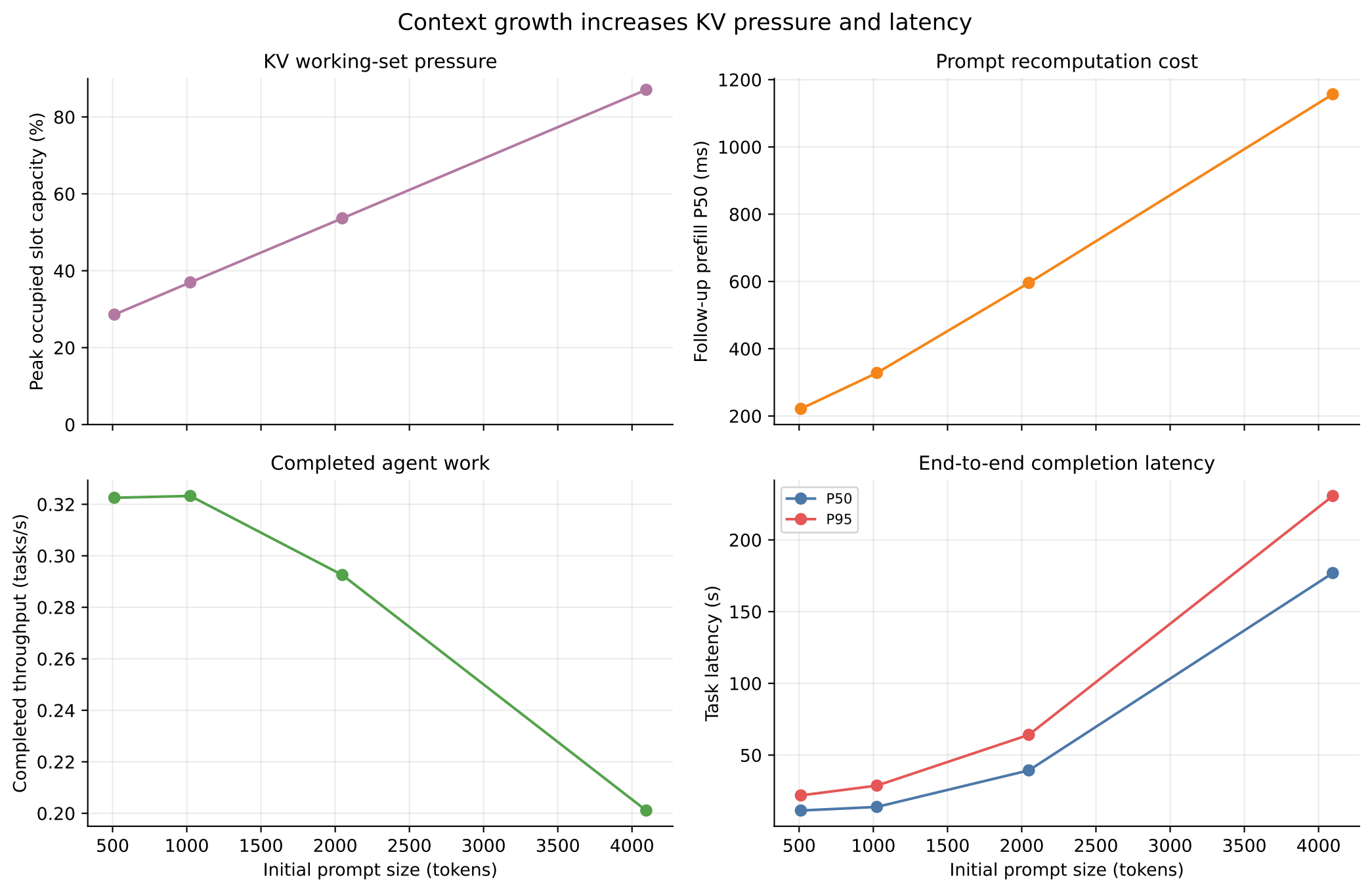
Context growth increases KV pressure and latency
KV working-set pressure
Peak occupied slot capacity (%)
0
20
40
60
80
Prompt recomputation cost
Follow-up prefill P50 (ms)
200
400
600
800
1000
1200
Completed agent work
Completed throughput (tasks/s)
0.20
0.22
0.24
0.26
0.28
0.30
0.32
500
1000
1500
2000
2500
3000
3500
4000
Initial prompt size (tokens)
End-to-end completion latency
P50
P95
Task latency (s)
50
100
150
200
500
1000
1500
2000
2500
3000
3500
4000
Initial prompt size (tokens)